

FOOD REMEDY API

Research: RE010

Prepared for: Food Remedy API

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SUMMARY

This report investigates the ongoing maintenance requirements for the food remedy application after its public release. The purpose of the research is to identify the technical, security, operational and compliance activities that Gopher industries should undertake to ensure that food remedy continues to operate reliably over time. Food Remedy is a mobile application that provides food related functionality such as product scanning, product information, allergen and dietary suitability information, user accounts, shopping list functionality and meal planning features. The application uses technologies including Expo, React Native, Firebase, Authentication, Cloud Firestore, Firebase Storage, SQLite and third party npm dependencies. These technologies are continuously updated, which means that the application will require ongoing maintenance rather than remaining unchanged after its initial release.

The research focuses on six major areas: Google Play Requirements, Apple App Store requirements, Application and SDK maintenance, security and dependency maintenance, backend monitoring, and maintenance scheduling. Google Play currently requires new applications and updates to target Android 16/API 36 or higher from 31 August 2026. Apple currently requires apps uploaded to the App Store Connect to be built with the Xcode 26 or later using the appropriate 26 SDKs.

The research recommends that Food Remedy adopt a proactive maintenance strategy. Critical security and production issues should be addressed immediately, dependencies and security should be reviewed at least monthly, platform and SDK requirements should be reviewed regularly, and comprehensive application testing should be performed before major releases.

This approach will reduce a major risk of security vulnerabilities, application incompatibility, store rejection, backend failures and poor use of experience.

INTRODUCTION

BACKGROUND:

The release of a mobile application will not show us the end of the software development life cycle. Once the application becomes fully publically available, the technical environment continues to change. Operating systems receive new versions, App Store introduces new requirements, development frameworks are updated, 3rd party packages become outdated and security vulnerabilities are discovered. For the Food Remedy, this is particularly important because the application depends on multiple external technologies and services. These include Expo and React Native for the mobile Application, Firebase for authentication and cloud services, SQLite for the local data and numerous npm packages.

For example, Expo maintains a relationship between the SDK versions and the React Native versions. Expo's documentation shows that Expo SDK 54 uses React Native 0.81, while the newer SDK releases use newer React Native versions. Expo also documents specific Andriod, iOS and Xcode requirements for each SDK version. Food Remedy needs a maintainence process that considers not only bugs in the application's own code but also changes occurring in the wider technical environment.

The research have the following objectives:

Investigate Google Play maintenance and publishing requirements.

Investigate Apple App store maintenance and publishing requirements.

Identify application, Expo and react native upgrade requirements.

Investigate dependency and security maintenance.

Investigate firebase and backend monitoring requirements.

To recommend an appropriate maintenance schedule.

Identify the risk associated with delayed maintenance and provide practical recommendations for maintaining the food remedy after release.

FOOD REMEDY APPLICATION MAINTAINENCE REQUIREMENTS

Food remedy use technologies such as Expo, react native, firebase authentication, cloud fire store, firebase storage, SQLite and npm dependencies. So, maintenance must cover both mobile application and backend service.

The main maintenance areas are:

Application and SDK updates.

App Store and Google Play compliance.

Security and dependency updates.

Firebase/backend testing.

Testing and bug fixing.

Monitoring changes to platform requirements.

The activities are important because changes to the operating systems, development frameworks or 3rd party services can affect the functionality of the application even when no changes are made to the application 's own source code.

GOOGLE PLAY AND APP STORE MAINTAINENCE

GOOGLE PLAY:

Food remedy will need to remain compliant with Google Play's technical and policy requirement. The important requirement is Android target API level. Google Play requires new applications and updates to target supported Android API levels, with the requirements increasing overtime. From August 31 2026, the new apps and updates must target Android 16 (API 36) or higher.

The team must check regularly the following:

Check the latest Google Play target API requirements.

Update the Android SDK or target API when required.

Check Google Play policy changes.

Keep the applications Data Safety and privacy information accurate.

Test the application on supported Android versions before releasing updates.

If we fail to maintain these requirements, this could prevent the application from submitting updates or cause compatibility problems with the newer Android devices.

APPLE APP STORE:

Apple also regularly changes its technical requirements. For example, the applications submitted to App Store connect from April 28 2026 must be built using Xcode 26 for later and the relevant SDKs.

Food remedy must therefore do the following:

Monitor Apple's submission requirements.

Keep Xcode and iOS versions supported.

Maintain compatibility between Expo, react native and iOS.

Test the application on supported iOS versions and devices.

Review App Store policies when releasing major updates.

Apple also demands the applications to remain functional and supported.

*Applications that no longer function as intended or are no longer actively supported may be removed from the App Store.

APPLICATION & SDK MAINTAINENCE

Food remedy uses Expo and react native, meaning the application will need to be maintained as these frameworks develop. Each Expo SDK IS associated with the particular react native version and has specific platform requirement.

Therefore, the team should not upgrade Expo or react native without checking compatibility with the rest of the project.

A suitable upgrade progress could be as follows:

Checking for a new Expo/react native release.

Review the release notes and breaking changes.

Check compatibility with Food Remedy dependencies.

Upgrade the required packages.

Run automated tests.

Test important application functions manually.

Test on both Android and iOS.

Release the update after successful testing.

The team must also avoid allowing the project to remain an outdated SDK version for long period, As large upgrades become more difficult when several versions are skipped.

SECURITY AND DEPENDENCY MAINTAINENCE

Food Remedy depends on a number of 3rd party npm packages. These dependencies can contain security vulnerabilities or become outdated over time. The development team must make sure to regularly run “npm audit” this allows to identify known vulnerabilities in installed dependencies. npm recommends reviewing vulnerabilities and determining whether updates or other actions are required or not.

The team should also:

Review dependencies regularly.

Apply security updates where appropriate.

Remove unnecessary dependencies.

Avoid using unsupported or abandoned packages.

Test the application after dependency updates.

Prioritise critical security vulnerabilities.

Updates should not be simply installed without testing because major dependency updates can introduce breaking changes.

FIREBASE & BACKEND MAINTAINENCE

Firebase is an important part of food remedy because it supports functionality such as authentication, firestore and storage. Therefore, backend maintenance is required in addition to mobile application maintenance. The team should also regularly monitor:

Firebase authentication errors and failures.

Firestore usage and errors.

Firebase storage usage.

Back end or API failures.

Unusual or unexpected activity.

Firebase SDK updates.

Firebase security rules should also be reviewed whenever new features or database structures are introduced. Firebase recommends using restrictive security rules and testing them before deployment. For example if the food remedy introduces a new fire soil collection, the team must verify that only the appropriate users or application components can access that data. Firebase also recommends monitoring backend services and using additional security measures such as App check where appropriate.

RECOMMENDED MAINTAINENCE SCHEDULE

Frequency	Maintenance activity	Application to Food Remedy
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Continuous / As required	Monitor critical production issues and security alerts	Monitor Firebase Authentication, Firestore, Storage and application errors. Critical failures should be investigated immediately. Firebase specifically recommends monitoring and alerting for backend services. (Firebase)
Monthly	Dependency and security review	Run <code>npm audit</code> , review vulnerable/outdated packages and apply appropriate security patches. npm classifies critical vulnerabilities as requiring immediate attention and high vulnerabilities as requiring action as quickly as possible. (npm Docs)
Monthly	Backend review	Review Firebase Authentication errors, Firestore usage, Storage usage, unusual activity and Security Rules changes.
Quarterly	SDK and framework review	Check Expo, React Native, Firebase SDKs and major npm dependencies for supported versions and known issues. Upgrade where necessary and test before deployment.
Quarterly	Security review	Review Firebase Security Rules, API configuration, authentication, permissions and App Check configuration. Firebase recommends App Check and restrictive security controls for supported services. (Firebase)
Quarterly	Platform compatibility review	Check current Android/iOS versions, Android target API requirements, Xcode requirements and Expo compatibility.
Before every major release	Full regression testing	Test login, registration, product scanning, product lookup, allergen information, shopping lists, meal planning, profiles and Firebase connectivity on both Android and iOS.
At least 3 months before a platform deadline	Platform upgrade planning	Begin migration when Google/Apple announces a major requirement. Google specifically recommends starting target API migration at least three months before the deadline. (Google Help)
Immediately	Critical security or production issue	Patch critical vulnerabilities, authentication failures, major crashes or backend outages rather than waiting for the next scheduled maintenance cycle.

The monthly or dependency review is particularly applicable to the food remedy because the project uses a large number of npm dependencies. “npm audit” Can identify non vulnerabilities in the dependency tree, allowing the team to determine whether updates or other remediation are required. The quarterly SDK review is appropriate because food remedy relies on Expo or react native

and it must remain compatible with changing Android and iOS requirements. Platform requirements can also have fixed deadlines. For example Google Play requires new application and updates to target Android 16 or API 36 from August 31 2026, while Apple App store requires submissions to use Xcode 26 or later with the relevant 26 SDKs as of April 28 2026. The 3 month preparation. Is also supported by Google's current guidance, which recommends the beginning target API migration at least 3 months before the deadline to allow time for compatibility testing and resolving problems.

RISKS OF DELAYED MAINTAINENCE

If maintainence activities are not performed regularly, Food Remedy could experience several risks:

Security Risks:

Outdated dependencies or incorrectly configured firebase security rules could introduce vulnerabilities and potentially expose application or user data.

Compatibility Risks:

Outdated Expo, react native, Android or iOS components could cause the application to stop working correctly on newer devices and operating systems.

Store Compliance risks:

Failure to meet Google Play or apple requirements could prevent new versions of food remedy from being submitted or distributed.

Backend Availability Risks:

without monitoring firebase and other backend services, authentication, database or storage problems may remain undetected for long.

Increased Technical Debt:

delaying updates can result in multiple outdated dependencies accumulating, making future upgrades more difficult and time consuming.

User Experience Risks:

Unresolved bugs, crashes or outdated functionality can reduce user satisfaction and confidence in the application.

RECOMMENDED APPROACH FOR FOOD REMEDY

Based on the research, food remedy should use a proactive maintenance approach rather than waiting until the problem occurs.

The team should do the following:

Monitor Google Play and apple requirements regularly.

Keep Expo, react native and other major dependencies reasonably up to date.

Perform monthly security and dependency checks.

Monitor firebase and backend services.

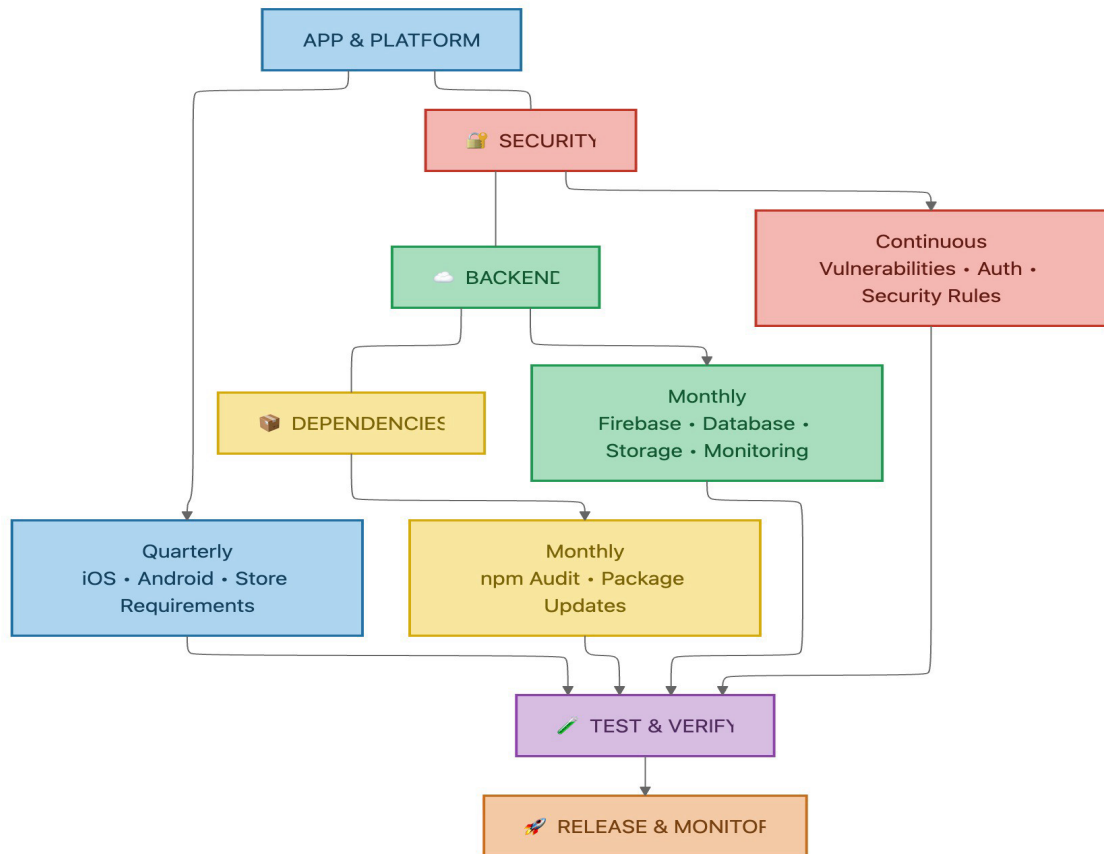
Review firebase security rules when functionality changes.

Test all major updates on both Android and iOS.

Address critical security and production issues immediately.

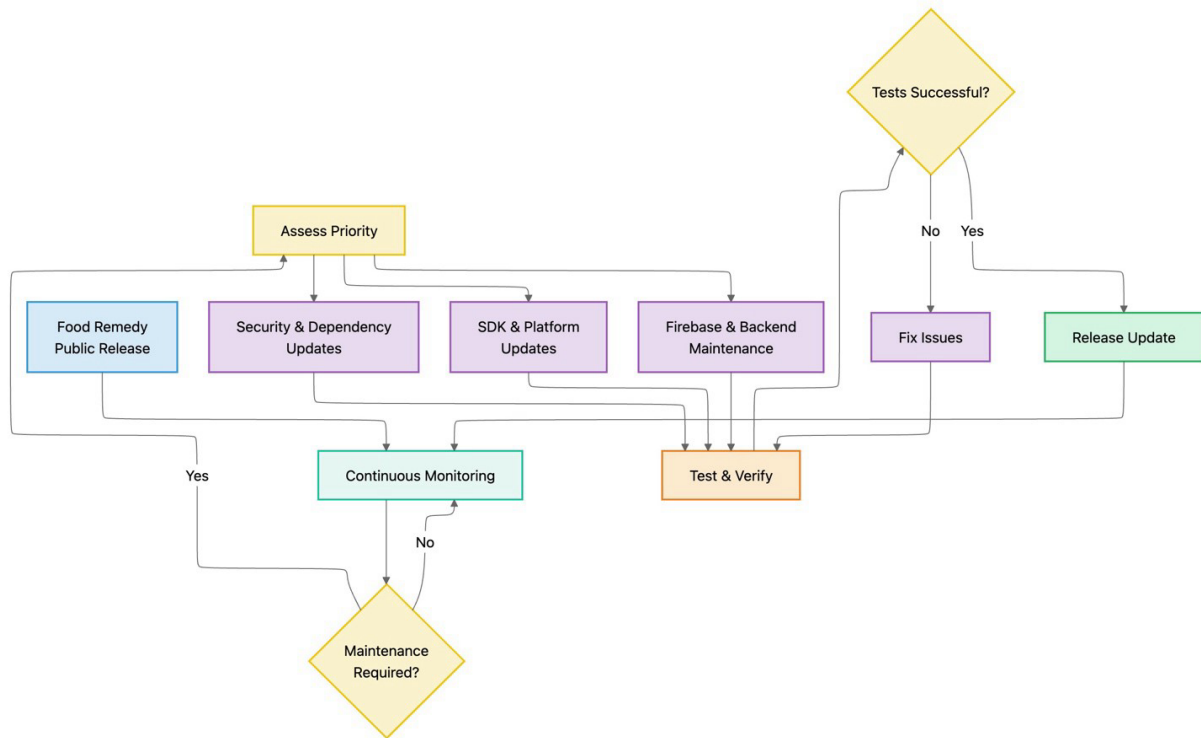
Keep maintenance activities documented so future team members can understand what has been updated and why.

This approach will help the food remedy remain secure, compatible, available and compliant after the public release.



CONCLUSION

Maintaining food remedy after release will require ongoing technical, security and operational work. The main areas requiring attention are App Store compliance, SDK and dependency updates commerce security fire base maintenance, monitoring and regular testing. A combination of monthly checks, quarterly technical reviews and immediate responses to critical issues provides practical maintenance strategy for the project and will reduce the risk of security vulnerabilities, compatibility issues and store compliant issues and service disruptions while helping ensure that the application continues to provide a reliable experience for users.



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